A Final Project Proposal on

Devanagari Handwritten Document Recognition Using OCR

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ABSTRACT

This project proposes a deep learning–based Optical Character Recognition (OCR) system for handwritten Nepali text written in the Devanagari script, with a focus on documents commonly found in government offices and scanned paper records. The system will employ Convolutional Neural Networks (CNNs) for visual feature extraction and Transformer-based models to capture contextual and sequential dependencies in handwritten text. The primary objective is to digitize handwritten Nepali government documents by converting them into machine-readable and editable Unicode text. A new labeled dataset of handwritten Nepali characters will be collected from students and representative government-style documents for model training and evaluation. A comparative analysis between CNN-based and Transformer-based approaches will be conducted using standard metrics such as Accuracy and Character Error Rate (CER) The proposed system will be implemented using Python and demonstrated through a simple web-based interface, contributing to improved accessibility, searchability, and digital preservation of Nepali handwritten records.

Keywords: Handwritten Character Recognition, Optical Character Recognition, Nepali Script, Devanagari, Deep Learning, Transformers, CNN.

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ACRONYMS

API	Application Programming Interface
BiLSTM	Bidirectional Long Short -Term Memory
CER	Character Error Rate
CRNN	Convolutional Recurrent Neural Network
CTC	Connectionist Temporal Classification
EDA	Exploratory Data Analysis
GPU	Graphics Processing Unit
HCR	Handwritten Character Recognition
LSTM	Long Short Term Memory
OCR	Optical Character Recognition
PDF	Portable Document Format
RNN	Recurrent Neural Network
UML	Unified Modeling Language

CHAPTER 1: INTRODUCTION

1.1 Background

The Government of Nepal continues to rely extensively on paper-based documentation across key sectors such as land registration, judiciary records, education, civil administration, and local municipalities. A large portion of these records are written in Nepali using the Devanagari script, either in handwritten or printed form. Consequently, the digitization of such documents into searchable, editable, and machine-readable text remains a major challenge for effective e-governance, digital archiving, and data-driven decision-making[1][2].

Optical Character Recognition (OCR) is a critical technology that enables the automatic conversion of scanned document images into digital text. While OCR systems for printed English and other high-resource languages have achieved high accuracy, OCR for Nepali Devanagari documents—especially handwritten ones—remains unreliable. This limitation arises from the structural complexity of the Devanagari script, which includes modifiers, compound characters, shirorekha connections, and visually similar symbols, as well as large variations in handwriting styles and document quality[5].

Traditional OCR approaches for Nepali have largely focused on isolated character recognition using handcrafted features or basic deep learning models such as Convolutional Neural Networks (CNNs). Although CNN-based models have shown promising performance in handwritten character recognition tasks, they struggle to capture sequential dependencies and contextual relationships between characters in real-world documents. Moreover, the absence of large, standardized, and diverse Nepali OCR datasets further limits their generalization capability[2][5].

Recent advancements in deep learning, particularly sequence-based models (CNN–RNN with CTC loss) and Transformer-based architectures, have significantly improved OCR performance in complex and low-resource scripts. Transformers, with their attention mechanisms, are capable of modeling long-range dependencies and contextual information, making them well-suited for full document-level text recognition. These developments motivate the exploration of Transformer-based OCR models for Nepali documents and their comparison with conventional deep learning approaches[6][7].

This project addresses a national-level challenge by proposing an end-to-end Nepali OCR system that converts scanned Nepali documents into machine-readable Nepali text using deep learning and Transformer-based models, along with a comparative performance analysis of these approaches.

1.2 Problem Statement

Despite ongoing digital transformation initiatives, Nepal faces significant difficulties in converting handwritten and scanned Nepali documents into usable digital text. Most existing OCR systems are optimized for printed text or non-Devanagari scripts and fail to accurately recognize Nepali text from real-world scanned documents, particularly handwritten records. As a result, a vast volume of government documents remains inaccessible in digital form[4].The lack of a reliable Nepali OCR system leads to heavy dependence on manual data entry, which is time-consuming, error-prone, and inefficient. Additionally, scanned Nepali documents are not searchable, not easily translatable, and cannot be effectively analyzed or integrated into digital workflows. While CNN-based handwritten character recognition models provide a partial solution, they are insufficient for handling full document recognition due to limited contextual understanding[8][9].

Therefore, there is a need for a robust OCR framework that leverages modern deep learning and Transformer-based architectures to accurately recognize Nepali Devanagari text from scanned documents. Furthermore, a systematic comparative

analysis is required to evaluate the effectiveness of traditional deep learning OCR models against Transformer-based approaches in the context of Nepali language documents.

1.3 Objectives

The main objective of this project includes:

i. To develop an end-to-end Optical Character Recognition (OCR) system for Nepali documents written in the Devanagari script using deep learning and Transformer-based models.

ii. To perform a comparative analysis between traditional deep learning–based OCR models and Transformer-based OCR models for recognizing printed and handwritten Nepali text.

1.4 Significance of the research

The Government of Nepal relies extensively on Nepali-language documents for administrative, legal, historical, and archival purposes. A large proportion of these documents exist only in handwritten or scanned formats, making them difficult to search, analyze, translate, and preserve digitally. The dependence on paper-based records significantly limits the effectiveness of e-governance initiatives and increases the risk of permanent data loss due to natural disasters, accidents, or social unrest. Recent incidents causing damage to government offices and physical records have further highlighted the vulnerability of traditional documentation systems and the urgent need for reliable digital preservation solutions[5][9].

This research is significant as it explores modern deep learning and Transformer-based OCR models for Nepali documents, moving beyond isolated character recognition toward full document-level text recognition. By conducting a comparative analysis between traditional deep learning approaches and Transformer-based architectures, the study provides valuable insights into the most suitable models for low-resource language OCR. The outcomes of this research can contribute to

improved document digitization, enhanced accessibility of Nepali textual data, and long-term digital preservation, supporting Nepal's e-governance, archival modernization, and digital transformation initiatives[3][4].

1.5 Scope and Limitation

This project focuses on the development of an Optical Character Recognition (OCR) system for Nepali documents written in the Devanagari script using deep learning and Transformer-based models. The scope of the research includes the recognition of printed and handwritten Nepali text from scanned images and PDF documents. The system aims to convert document images into machine-readable Nepali text, thereby improving document accessibility, searchability, and digital usability.The study includes the implementation and evaluation of multiple OCR architectures, including deep learning–based models (such as CNN and CNN–RNN with CTC loss) and Transformer-based models. A comparative performance analysis is conducted using standard evaluation metrics such as accuracy and Character Error Rate (CER).The datasets used in this research consist of scanned Nepali documents and handwritten samples collected from students and representative writing styles to reflect real-world document conditions[3][6][7].

Despite the potential benefits of the proposed OCR system, certain limitations must be acknowledged. Recognition performance may be affected by poor image quality, document noise, skew, low resolution, and extreme variations in handwriting styles. The accuracy of the OCR models is highly dependent on the quality, diversity, and size of the available training data; limited or biased datasets may reduce capabilityAdditionally, the system may face challenges when processing severely degraded historical documents, complex layouts, or highly cursive handwriting. While Transformer-based models improve contextual understanding, they require

higher computational resources compared to traditional deep learning models, which may limit deployment in resource-constrained environments. In such cases, partial manual verification or correction may still be required[9][10].

CHAPTER 2 : LITERATURE REVIEW

2.1 Overview of Existing Systems

Research on handwritten and printed character recognition for Devanagari-script languages has received considerable attention due to the complexity of the script, which includes modifiers, compound characters, and visually similar symbols. Early studies mainly focused on handcrafted feature extraction techniques combined with traditional classifiers. Pal and Chanda proposed an offline handwritten Devanagari character recognition system using a hybrid combination of Support Vector Machines (SVM) and Modified Quadratic Discriminant Function (MQDF). By using gradient and curvature-based features, their approach achieved high recognition accuracy; however, confusion between similar-looking characters remained a major limitation [1][2].

Several works have specifically aimed at improving Nepali OCR performance using hybrid and holistic recognition strategies. Pant and Bal introduced a hybrid OCR approach that combined holistic word-level recognition with character-level recognition, resulting in nearly a 10% improvement in OCR accuracy. Their system achieved around 90% accuracy at the word level and 78.87% accuracy at the character level. Despite the improvement, segmentation issues such as over-segmentation and under-segmentation were identified as key challenges, especially for compound and shadow characters. The authors suggested recognition-driven segmentation and better training of compound characters as future solutions[3][4].

Recent research has expanded beyond character recognition to document processing and natural language understanding tasks. Thapa et al. conducted a shared task on Natural Language Understanding (NLU) for Devanagari-script languages, focusing on language identification and hate speech detection. Their results showed that transformer-based models with multilingual embeddings achieved near-perfect performance in language identification, although challenges such as class imbalance and limited labeled data persisted. Similarly, Paudel et al. compared PDF parsing and

OCR techniques for extracting Nepali text from PDFs and found that OCR-based approaches, particularly PyTesseract, were more reliable for both scanned and image-based documents. These studies highlight the importance of robust OCR and language-aware models for real-world Nepali and Devanagari text processing[5][6].

2.2 Comparison of Features

The following table compares the key features of the existing crop recommendation systems, highlighting their strengths, weaknesses, and notable features.

Table 2-1: Comparison of features

Authors	**Description**	**Features**	**Strengths**	**Weakness**
Pal & Chanda [1]	Offline handwritten Devanagari character recognition	Gradient and Curvature Features	High accuracy using SVM + MQDF	Confusion between similar characters
Thapa et al. [3]	NLU for Devanagiri-script languages	Multilingual embeddings, transformer features	Excellent in language identification	Class imbalance and limited labeled data
De Campos et al [4]	Character recognition in natural images using real and synthetic data	Geometric blur, shape Context, font based features	Shape-based features perform well; synthetic data reduces data cost	Low accuracy in complex backgrounds
Pal & Singh	Handwritten	Fourier descriptors,	Good recognition,	Increased training

[6]	English character recognition	neural networks	accuracy	time with larger networks
Pant & Bal [2]	Hybrid Nepali OCR using word-level and character-level recognition	Holistic word features, character-level features, multiple classifiers	Nearly 10% OCR accuracy improvement; word-level accuracy ~90%	Performance affected by segmentation issues and limited compound character training

2.3 Gaps in Existing Systems

2.3.1 Lack of Generalization : Most existing OCR and character recognition systems are trained on limited datasets and perform well only for specific scripts, fonts, or languages. Models trained on controlled datasets often fail to generalize to real-world data such as handwritten text, scanned documents, or natural scene images. This limits their applicability across different Devanagari-script languages like Nepali, Hindi, and Marathi.

2.3.2 Segmentation Errors: A major challenge in Devanagari OCR systems is segmentation. Over-segmentation and under-segmentation frequently occur due to modifiers, compound characters, and touching symbols. These segmentation errors significantly reduce recognition accuracy, especially in handwritten and degraded documents.

2.3.3 Limited Handling of Complex Characters: Many existing systems struggle with conjunct and compound characters, which are common in Nepali and other Devanagari-based languages. Most models focus on basic characters and do notsufficiently address shadow characters or ligatures, leading to misclassification and reduced performance.

2.3.4 Dependence on Large Labeled Datasets: Modern deep learning and transformer-based models require large amounts of labeled data for effective training.

However, annotated datasets for Nepali and low-resource Devanagari languages are limited, making it difficult to build robust and scalable recognition systems.

2.3.5 High Computational Requirements: Advanced OCR and deep learning models often require high computational resources and powerful hardware. This makes them less suitable for deployment in small-scale, educational, or resource-constrained environments, particularly in developing regions.

2.4 Significance of Proposed Work

2.4.1 Scalable and Cost-Effective Design: The proposed system prioritizes accessibility using lightweight frameworks like Streamlit. It integrates real-time stock APIs and optimized algorithms to reduce memory and computational load, ensuring suitability for rural or educational settings.

2.4.2 Improved Accuracy through Hybrid Recognition: By combining character-level and holistic recognition approaches, the proposed work aims to improve overall OCR accuracy for Nepali text. This hybrid strategy is expected to handle segmentation issues more effectively and improve recognition of compound and complex characters.

CHAPTER 3: METHODOLOGY

This chapter presents the proposed methodology for the design and implementation of an end-to-end Optical Character Recognition (OCR) system for Nepali documents written in the Devanagari script. The proposed methodology will integrate traditional deep learning–based OCR architectures and modern Transformer-based models to recognize printed and handwritten Nepali text from scanned images and PDF documents. A comparative analysis will be conducted to evaluate the performance, robustness, and suitability of these approaches for low-resource language OCR. The overall methodology will follow a structured pipeline consisting of data collection, data preprocessing, model design, model training, evaluation, and comparative analysis[3][6][7].

3.1 Data Collection

The dataset for this research will consist of scanned Nepali documents and handwritten Nepali text samples written in the Devanagari script. Data will be collected from multiple sources to ensure diversity in writing styles, document layouts, and image quality[8]. The proposed data sources include:

i. Custom handwritten samples collected from students and individuals of different age groups to capture natural handwriting variations.

ii. Scanned handwritten documents such as forms, notes, and government-style records to simulate real-world conditions.

iii. Publicly available Devanagari datasets used for benchmarking and pretraining where applicable.

3.2 Data Preprocessing

Data preprocessing will be a critical step in the proposed OCR system, as scanned documents often contain noise, skew, uneven illumination, and background artifacts. A standardized preprocessing pipeline will be applied to enhance image quality and ensure compatibility with deep learning and Transformer-based models[3].

3.2.1 Image Enhancement and Cleaning

The following preprocessing operations will be applied:

i. Conversion of RGB images to grayscale

ii. Noise removal using Gaussian and median filtering

iii. Contrast enhancement and adaptive thresholding

3.2.2 Skew Correction and Normalization

To improve recognition accuracy, the following normalization steps will be performed:

i. Detection and correction of document skew

ii. Resizing images to a fixed resolution suitable for model input

iii. Pixel value normalization to stabilize model training

3.2.3 Data Augmentation

To improve model generalization and reduce overfitting, data augmentation processes such as rotation, scaling, translation, and slight distortion will be applied to the training dataset. These transformations will simulate real-world scanning variations while preserving textual content.

3.3 OCR Model Architecture

The OCR system is designed using two distinct model pipelines to enable comparative analysis.

3.3.1 Deep Learning–Based OCR Model (CNN–RNN–CTC) :

The first pipeline employs a Convolutional Recurrent Neural Network (CRNN) architecture. Convolutional Neural Networks (CNNs) extract visual features from text images, while Bidirectional Long Short-Term Memory (BiLSTM) layers model the sequential nature of Nepali text. Connectionist Temporal Classification (CTC) loss is used to align predicted character sequences with ground truth text without explicit character segmentation. This architecture serves as the baseline deep learning OCR model[2][3].

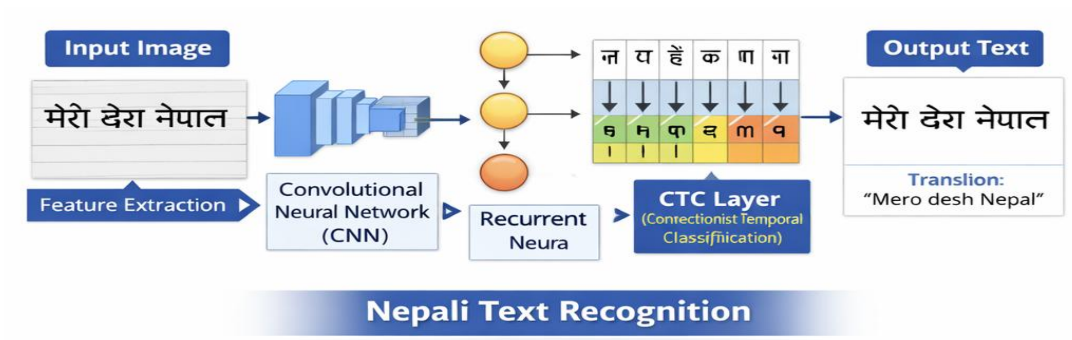

Figure 3-1:Deep Learning–Based OCR Architecture (CNN–RNN–CTC)

3.3.2 Transformer-Based OCR Model :

The second pipeline utilizes a Transformer-based OCR architecture. Visual features are extracted using a CNN backbone and converted into patch embeddings. These embeddings are processed by a Transformer encoder–decoder framework that models long-range dependencies and contextual relationships between characters and words. Attention mechanisms enable the model to capture complex Devanagari script structures such as modifiers, compound characters, and shirorekha connections[2][3].

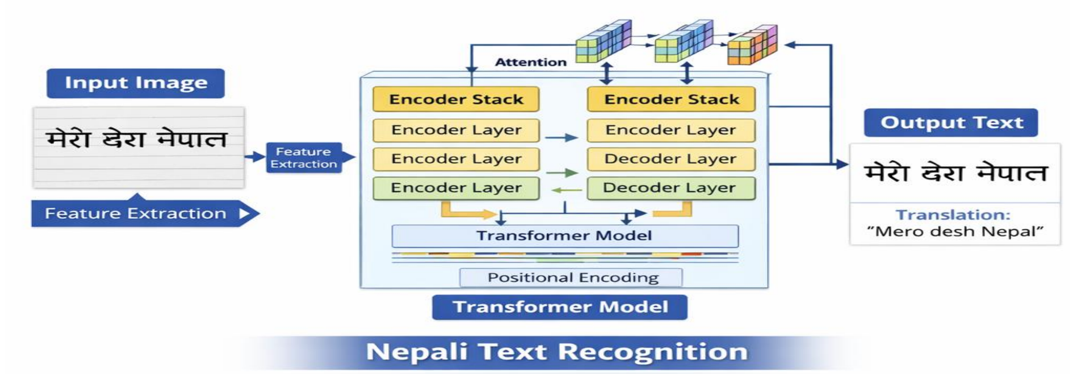

Figure 3-2:Transformer-Based OCR Architecture

3.4 Model Training

Model training will be conducted using supervised learning techniques to enable accurate recognition of Nepali Devanagari text. Separate training strategies will be applied for the deep learning–based OCR model and the Transformer-based OCR model to ensure fair comparison and effective learning[1].

3.4.1 Dataset Splitting

The complete dataset will be divided into training, validation, and testing sets. The training set will be used to learn model parameters, the validation set will support hyperparameter tuning and performance monitoring, and the testing set will be reserved for final evaluation[3][4].

3.4.2 Training of CNN–RNN–CTC Model

The CNN–RNN–CTC (CRNN) model will be trained to recognize sequential Nepali text without explicit character segmentation. The CNN component will extract visual features such as strokes and character shapes, while Bidirectional LSTM layers will

model character sequences. Connectionist Temporal Classification (CTC) loss will be used to align predicted sequences with ground-truth Nepali text. The model will be optimized using the Adam optimizer with learning rate scheduling, along with regularization techniques such as dropout and early stopping[6][7].

3.4.3 Training of Transformer-Based OCR Model

The Transformer-based OCR model will follow an encoder–decoder training approach. Visual features extracted using a CNN or Vision Transformer backbone will be processed by transformer encoders to capture global contextual dependencies. The decoder will generate Nepali text sequences using attention mechanisms. Cross-entropy loss with teacher forcing will be applied during training. Pretrained weights will be used where applicable, and the model will be fine-tuned using the AdamW optimizer to improve convergence and generalization[3][7].

3.5 Inference and Text Generation

During inference, scanned Nepali documents will be processed through the same preprocessing pipeline. Text regions will be detected and passed to the recognition models, which will generate Unicode Nepali text sequences. The recognized text will be post-processed to correct minor recognition errors and ensure proper Unicode formatting[4].

3.6 Comparative Analysis

A comparative analysis will be conducted between the deep learning–based OCR model and the Transformer-based OCR model. Performance will be evaluated using standard OCR metrics such as Accuracy, Character Error Rate (CER), and Word Error Rate (WER). Computational complexity, training time, and robustness to handwriting variations will also be analyzed[2][3].

3.7 Model Evaluation

The performance of the proposed OCR system will be evaluated using both quantitative and qualitative methods.

3.7.1 Quantitative Evaluation Metrics

Quantitative evaluation metrics are numerical measures used to objectively access the performance,accuracy and effectiveness of a model or a system by comparing its outputs against known ground truth or expected results.The common evaluation metrics are accuracy, CER,WER,etc.

i. **Accuracy:**It measures how many predictions are completely correct out of all predictions[7].

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total predictions}}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- **TP (True Positive):** Correctly predicted positive
- **TN (True Negative):** Correctly predicted negative
- **FP (False Positive):** Wrongly predicted positive
- **FN (False Negative):** Wrongly predicted negative

ii. **Character Error Rate (CER):**CER measures how many character-level errors are made in text prediction (OCR, ASR)[7].

$$\text{CER} = \frac{S + D + I}{N}$$

Where:

- S = Substitutions (wrong characters in place of correct ones)
- D = Deletions (missing characters)
- I = Insertions (extra characters added)
- N = Total number of characters in the reference text

3.7.2 Qualitative Evaluation

Qualitative evaluation will involve visual inspection of recognized text outputs to assess readability and contextual correctness, particularly for handwritten Nepali documents. Errors related to modifiers, compound characters, and shirorekha connections will be analyzed in detail.

3.8 Experimental Environment

All experiments will be conducted in a GPU-enabled environment such as GoogleColab to ensure efficient training and reproducibility of both deep learning and Transformer-based OCR models.

3.9 Tools and Technology Stack

The proposed system will utilize the following tools and technologies:

i. Programming Language: Python

ii. Image Processing: OpenCV

iii. Deep Learning Frameworks: TensorFlow/Keras or PyTorch

iv. OCR Models: CRNN (CNN–RNN–CTC) and Transformer-based OCR models

v. Annotation Tools: LabelImg / Roboflow

vi. Frontend Interface: Streamlit

3.10 System Design

In the system design phase, system modeling will be carried out to provide a structured and visual representation of the proposed Optical Character Recognition (OCR) system for Nepali documents written in the Devanagari script. System modeling diagrams will be used to describe the overall system architecture,data flow,functional interactions, and operational behavior of the OCR system. These diagrams include the Context Diagram, Data Flow Diagram (DFD), Use Case Diagram, Entity Relationship (ER) Diagram, and Sequence Diagram. Each model will highlight a specific aspect of the system, such as external interactions, internal processing, data storage, and user involvement. Collectively, these diagrams will help in clearly defining system requirements and guiding the effective implementation of the proposed Nepali OCR system[5][8].

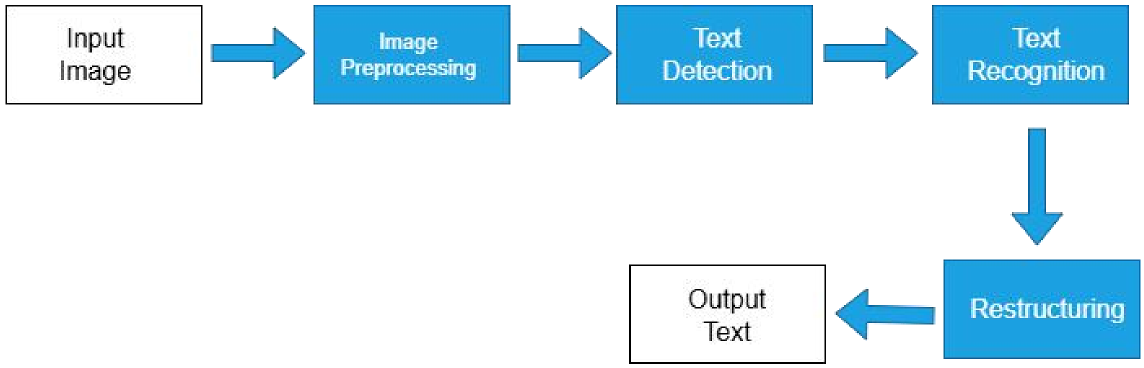

Figure 3-3:Schematic diagram of the proposed research framework

3.10.1 Context Diagram

The context diagram will provide a high-level overview of the proposed Nepali OCR system by illustrating its interaction with external entities. The primary external entity will be the user, who provides scanned Nepali documents or handwritten images as input to the system. The OCR system will process these inputs and produce machine-readable Unicode Nepali text as output. This diagram will define the system boundary and clearly show the input–output relationship without exposing internal processing details[7].The context diagram is shown in the figure below:

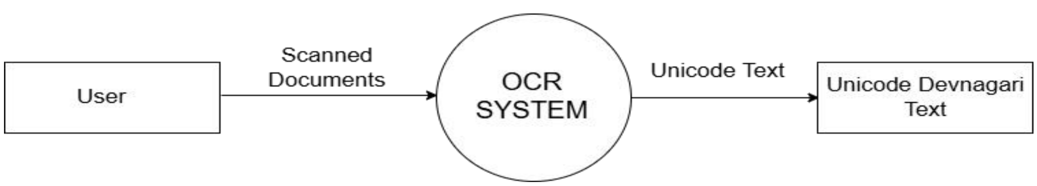

Figure 3-4: Context Diagram

3.10.2 Data Flow Diagram (DFD – Level 1

The Level 1 Data Flow Diagram will represent how data flows through the internal components of the Nepali OCR system. It will illustrate key processes such as image preprocessing, text detection, text recognition using deep learning and Transformer-based models, and text post-processing. The diagram will also include data stores such as annotated datasets, trained OCR models, and recognized text outputs. This model will help visualize how raw scanned documents are transformed into structured digital Nepali text[8][9].The DFD level 1 diagram is shown below:

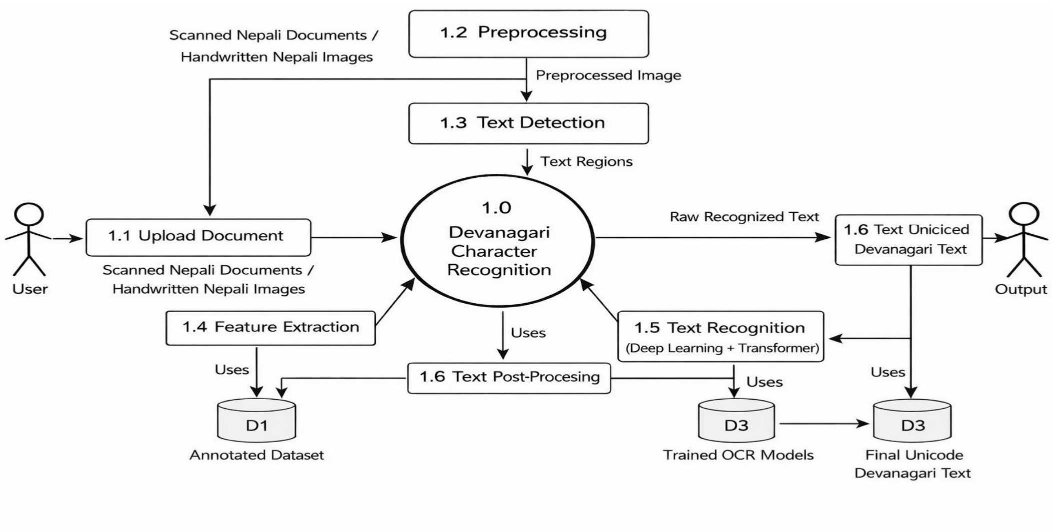

Figure 3-5: Data Flow Diagram (DFD-Level1)

3.10.3 Use Case Diagram

The use case diagram will describe the functional behavior of the system from the user's perspective. The primary actor will be the user, who interacts with the OCR system. The main use cases will include uploading scanned Nepali documents, preprocessing images, performing OCR, viewing recognized Nepali text, and downloading the output. This diagram will ensure that all essential system functionalities are clearly defined and aligned with user requirements[1][4].The use case diagram is shown below:

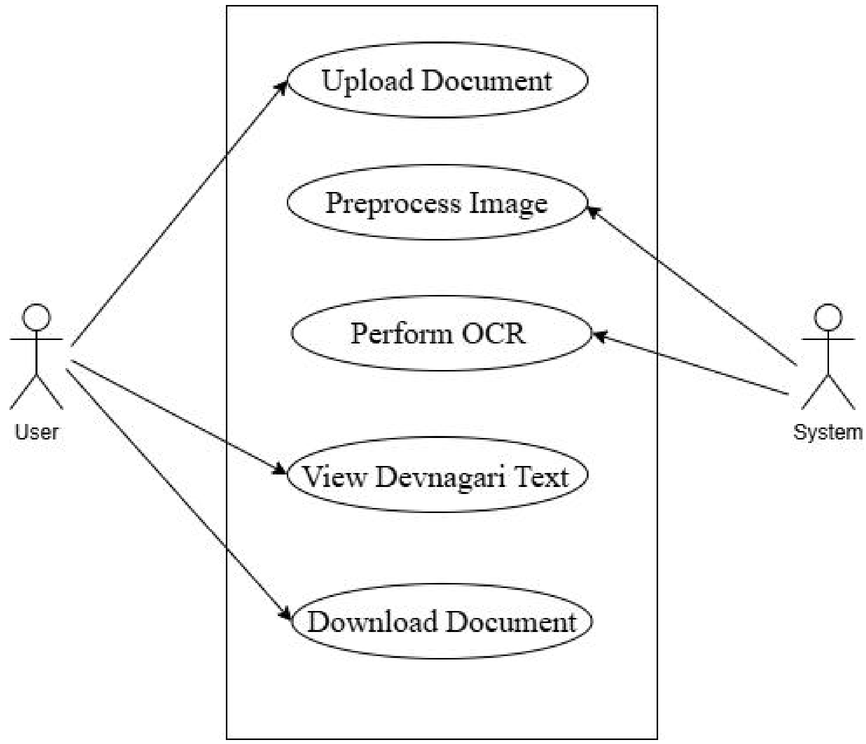

Figure 3-6: Use Case Diagram

3.10.4 Entity Relationship (ER) Diagram

The Entity Relationship (ER) diagram will model the logical data structure of the OCR system. Key entities will include Document Image, Text Region, Recognized Text, OCR Model, and Evaluation Results. Relationships will define how document images are linked to extracted text regions and recognized Nepali text outputs. This diagram will support efficient data organization and database design for storing documents, annotations, and OCR results[1][4].The entity relationship diagram is shown below:

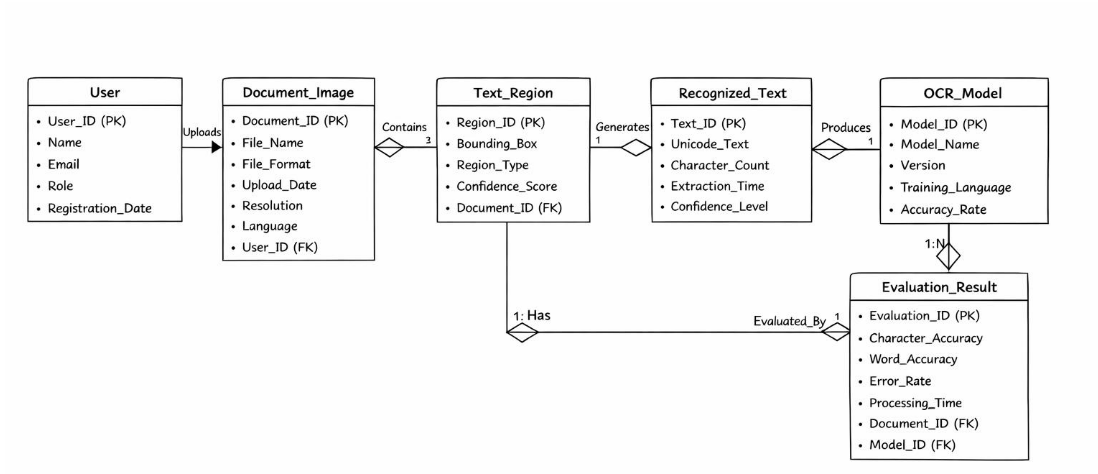

Figure 3-7: Entity Relationship (ER) Diagram

3.10.5 Sequence Diagram

The sequence diagram will illustrate the time-ordered interaction between system components during OCR processing. It will show how the user uploads a scanned Nepali document, which is then passed through preprocessing, text detection, and text recognition modules. The recognized Nepali text will then be post-processed and returned to the user. This diagram will clearly demonstrate the operational flow of the system from input submission to final OCR output[1].The sequence diagram is shown in the figure below:

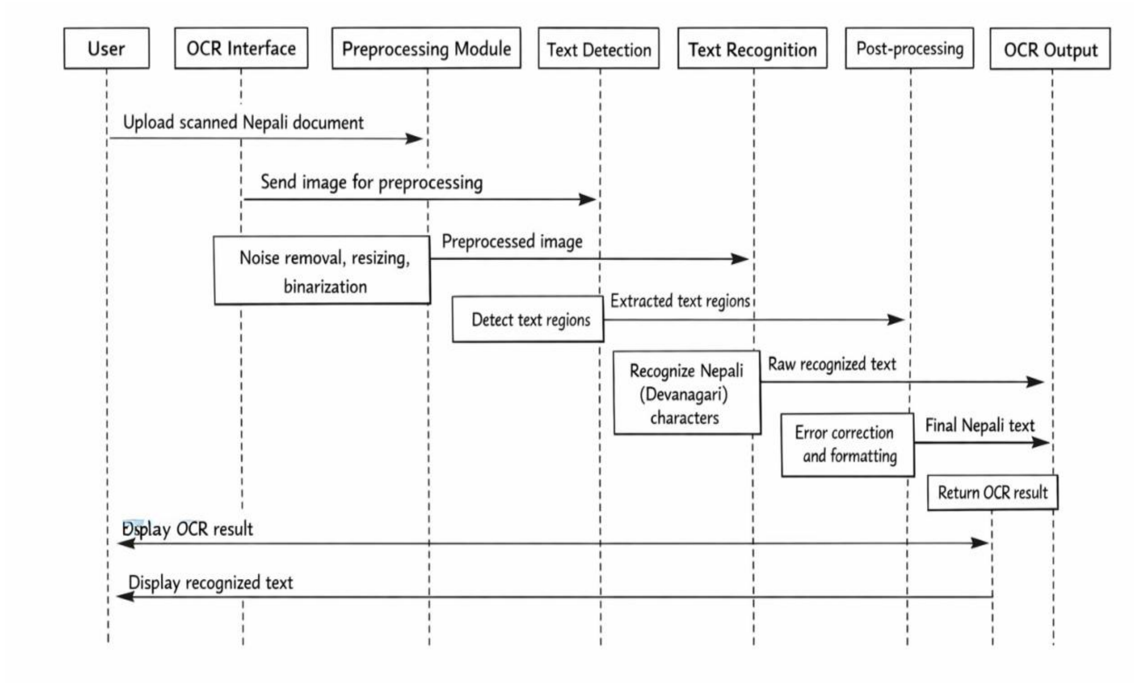

Figure 3-8: Sequence Diagram

CHAPTER 4: EXPECTED OUTCOME

The proposed research is expected to result in the development of a deep learning–based Optical Character Recognition (OCR) system capable of recognizing handwritten and scanned Nepali text written in the Devanagari script and converting it into machine-readable Unicode format. By employing both traditional deep learning approaches and Transformer-based models, the system is anticipated to achieve improved recognition performance for complex Devanagari characters, including modifiers, compound characters, and shirorekha connections. The study is expected to provide a comparative performance analysis using standard evaluation metrics such as Accuracy, Character Error Rate (CER),etc. offering insights into the effectiveness of different OCR architectures for low-resource languages. The outcomes of this research are expected to contribute to Nepali OCR literature and support practical applications in document digitization, digital archiving, and e-governance initiatives[3][4][9].

CHAPTER 5: CONCLUSION

This project proposes the development of a deep learning–based Optical Character Recognition (OCR) system for Nepali documents written in the Devanagari script. The primary objective of the proposed system is to accurately recognize handwritten and scanned Nepali text and convert it into machine-readable Unicode format. The system is designed to leverage modern deep learning architectures, including CNN–RNN–CTC models and Transformer-based OCR approaches, to address the challenges posed by complex Devanagari character structures, modifiers, and handwriting variations[7].

The proposed OCR framework will incorporate systematic data collection, preprocessing, model training, and evaluation strategies to ensure robustness and generalization. By performing a comparative analysis between traditional deep learning–based OCR models and Transformer-based models, the study aims to identify the most effective approach for Nepali text recognition in low-resource settings. Standard evaluation metrics such as accuracy, Character Error Rate (CER), and Word Error Rate (WER) will be used to assess system performance[4].

The successful implementation of the proposed system has the potential to significantly improve the digitization and accessibility of Nepali documents, particularly handwritten government and archival records. This research may contribute to e-governance initiatives, digital preservation efforts, and future advancements in Nepali language processing. Furthermore, the proposed work can serve as a foundation for extended research in areas such as full document OCR, Nepali text translation, and intelligent document analysis systems.

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Appendices A: Gantt chart

The timeline for completing this project may differ based on several factors, including the project's complexity, the scope of tasks, resource availability,etc.. Below is the Gantt Chart illustrating the work schedule:

Table A-1 Gantt Chart

S.N	Activities	January				February				March			
		1	2	3	4	1	2	3	4	1	2	3	4
1	Literature Review												
2	Preparation of Proposal												
3	Proposal Defense												
4	Coding Initialization												
5	Midterm Presentation												
6	Coding Continuation												
7	Report Preparation												